Amit Tiwari

EDUCATION

IIT BOMBAY
MS(R) in AI and Data Science
2025 - ongoing | Bombay, India
CPI: 9.65/10.0 (2 semesters)

IIT GUWAHATI
BTech. in Electronics and Electrical Engineering
2017 - 2021 | Assam, India
CPI: 9.09/10.0

RESEARCH FOCUS

BROAD DOMAINS:
MLLM ASR Accuracy, LLM Reasoning capabilities, RL post training , Inference Efficiency

ADVISOR:
Prof. Preethi Jyothi, CSALT Lab, IITB

KEY COURSES

GRADUATE
Foundations of Machine Learning
Statistical Signal Analysis
Algorithms and Complexity
Optimization
Matrix Computations
Efficient AI
Theoretical Machine Learning

UNDERGRADUATE
Digital Circuits and Microprocessors
Digital Signal Processing
Fundamentals of VLSI CAD

SKILLS

PROGRAMMING
Python • C • C++

AI / ML STACK
PyTorch • Hugging Face • scikit-learn

TOOLS
Git • Linux/Bash • LaTeX

SIMULATION
MATLAB • Simulink • LTSpice

ACHIEVEMENTS

UNDERGRADUATE
Awarded Institute Merit Scholarship for Academic Year 2020

PROJECTS

AUTHORITY AND INJECTION RESISTANCE IN CODING AGENTS
Red-teaming GPT-5.5 coding agent (Part of Solumn.ai coding assignment problem)
Oct 2026
• Built 5 prompt-injection scenarios with oracle controls and an external audit to test whether an agent treats untrusted content as authority and executes a malicious action in a sandboxed environment.
• **Review-bot injections** caused unauthorized waivers in 12/12 runs; the other 16 trials resisted all injections.

HIDDEN REASONING IN LLM SCRATCHPADS Extending Brauer et al., 2026 on Qwen2.5-7B-Instruct (Part of MATS application)
Sep 2026
• Corrupted scratchpad steps in 200 puzzles: answers followed the error 94% (English) vs 73% (symbolic), with 0/400 self-corrections.
• Probing and activation patching showed filler tokens carry operands but no computation (patching recovery 0.00–0.01).

REPLICATING DOUBLE DESCENT
Prof Abir De, IIT Bombay
Aug 2025 – Nov 2025
• Successfully demonstrated double descent phenomena on CIFAR-10 and MNIST datasets using Random Fourier Features. Original paper: Belkin et al., 2019
• Extended experiments to include FCNN, CNN, and ResNet architectures

SOLAR POWERED BATTERY CHARGING SYSTEM
Prof Ravindranath Adda, IIT Guwahati
Sep 2020 – May 2021
• Designed a charging solution and control system for electric vehicles, implementing DC-DC power electronics algorithms.
• Implemented Perturb and Observation algorithms in MATLAB and developed battery health preservation safety logic.

WORK EXPERIENCE

TEXAS INSTRUMENTS | Validation Engineer
Sep 2022 - Sep 2023 | Bengaluru, India
• Developed robust Python automation scripts and pipelines to streamline testing and characterization workflows for high-frequency RF devices.
• Automated test execution and data collection for AFE80XX high-speed converters.
• Designed impedance matching networks for high-frequency transceiver systems.

JAGUAR LAND ROVER INDIA | Validation Trainee, Powertrain
Jul 2021 - Jul 2022 | Bengaluru, India
• Developed custom GUI applications using Python's Tkinter library to automate inputs and streamline Model-in-the-Loop (MiL) testing for powertrain modules.
• Built automated test execution scripts and pipelines for PHEV validation using AVL PUMA software.

ONLINE TRAININGS

ECARS2X: ELECTRIC CARS: TECHNOLOGY Jul 2021
• Completed edX certification: `https://courses.edx.org/certificates/9aa0a1e075234827a7b1e22c9ce92604`